HANNAH MANOJ

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EDUCATION

Bachelor's of Science in Computer Science (HONS) **Expected Graduation, May 2027**
Leeds Beckett University, Leeds, England
Concentrations: Data Science, Object-Oriented Programming, Computing Systems (Linux), Applied Machine Learning, Database Systems, Object-Oriented Programming (Java), Software Systems Development, Operating Systems (Linux), Applied Machine Learning, Project Management, Computer Networks & Internet Fundamentals

EXPERIENCE

Business Intelligence Engineer Intern, Amazon, London, England **June 2025 - Present**
12 month placement
Designed and deployed an end-to-end **automated ETL pipeline** using AWS CDK, replacing a manual reporting workflow
Built and optimised **AWS Glue (PySpark)** jobs, reducing processing time from **80–100 minutes to ~20 minutes (~75% improvement)**
Implemented **event-driven architecture** using Amazon S3, EventBridge, Lambda, and Glue Crawlers to automate schema updates and downstream workflows
Reduced load on Amazon Redshift by migrating complex SQL transformations to scalable PySpark-based processing
Developed **incremental data refresh pipelines** and dashboards in Amazon QuickSight, enabling near real-time reporting for stakeholders
Wrote and optimised **complex SQL queries** on large-scale datasets to support business-critical reporting and analytics. Collaborated with cross-functional teams (engineering, product, operations) to translate business requirements into scalable data solutions Ensured **data quality and reliability** through validation checks, monitoring, and performance optimisation

PROJECTS

MOVIE RECOMMENDATION SYSTEM | GITHUB LINK: MOVIE-RECS
Tools Used: *Python, Pandas, Scikit learn, Machine learning, Jupyter Notebook*
Created an interactive movie recommendation system where you can type the name of a movie into an input box, and instantly get recommendations for other movies you might like.

SMART DOOR SECURITY SYSTEM (Full-Stack IoT Security Platform)
Stack: React Native, Python, TypeScript, NodeFlask, OpenCV, Firebase, Raspberry Pi, Ngrok - Built a full-stack smart security system combining facial recognition and fingerprint authentication to automate secure door access. Developed a **cross-platform React Native app** enabling users to register (OTP-based 2FA), manage access, view live camera feeds, and control the lock remotely. Implemented a **Flask backend on Raspberry Pi**, integrating camera, motion sensor, and solenoid lock for real-time detection and access control. Designed and shipped key user flows including onboarding, face registration, live monitoring, and alert notifications for recognised/unrecognised users. Enabled secure remote access by exposing local services via Ngrok, optimising video streaming performance for mobile devices.

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy, Scikit-learn), Java, JavaScript, SQL, React, Next JS, REST APIs, Node.js, API Design, Algorithms & Data Structures, OOP, OpenCV, Git
Data & Analytics: Data Mining, Data Pipelines, Power BI, Jupyter, ETL Processing, SQL optimisation, Data preprocessing, Model pipelines
Cloud & Infrastructure: AWS (CDK), Cloud Computing, Serverless Architecture, Docker, GIT, CI/CD, DNS resolution, HTTP/HTTPS protocols
Machine Learning & AI: LLMs, RAG, NLP, Computer Vision, Object detection

CERTIFICATIONS - AWS Certified AI Practitioner